

INTRODUCTION

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PROBLEM STATEMENT

Utility Creation from Waste: Developing functional objects or materials from institutional waste

Sorting and Segregating Waste



Materials

Non-biodegradable materials such as plastics, glass, and paper are collected and sorted.



Quality Control

Materials are inspected for contaminants to ensure the integrity of the final product.

Shredding and Mixing

Sorted materials are shredded into small, uniform particles, creating a consistent mixture.

The shredded materials are mixed with a binding agent to enhance cohesion and strength.

Molding and Curing

The mixed material is compressed and molded into brick shapes, using specialized equipment.

The molded bricks are cured, allowing the binding agent to solidify and the bricks to harden.



Novelty of Eco-Bricks



Environmental Sustainability

Reduces landfill waste by giving a second life to discarded materials.



Cost-Effectiveness

Provides an affordable alternative to traditional building materials.



Community Empowerment

Creates opportunities for local communities to build and improve their infrastructure.

100

Schools

Built or renovated using eco-bricks, providing sustainable learning environments.

5000

Homes

Constructed with eco-bricks, providing affordable housing for low-income families.

Market value and competitors.

The **global eco-bricks market** was valued at approximately **USD 25 million in 2023** and is projected to reach **USD 32 million by 2030**, growing at a CAGR of **3.9%** from 2024 to 2030.

Key Competitors in the Eco-bricks Market:

- **Green Antz Builders:** Specializes in producing eco-bricks and promoting eco-friendly construction materials.
- **Eco-bricks:** Focuses on creating and distributing eco-bricks for various applications.
- **The Plastic Bank:** Aims to reduce ocean plastic by turning waste into currency, supporting eco-brick initiatives.



Future Potential

Eco-bricks offer a promising solution to waste management and affordable construction. With ongoing research and development, the potential for this technology to transform construction practices and create a more sustainable future is vast.

Drivers

In the United States, **Nathan Gray** is the project lead for **Eco-Brick US**. He also founded the House Factory Foundation and Plastic Recycled, working to promote sustainability through eco-bricking.

The **Global Eco-brick Alliance (GEA)** is a leading organization promoting the eco-brick movement worldwide. Co-founded by **Russell Maier**, who serves as the lead developer, the GEA focuses on spreading eco-bricking practices and developing platforms like GoBrik to support the global plastic transition movement.

Barriers

Lack of Awareness: Limited knowledge or understanding of the concept or initiative, leading to resistance or indifference.

Financial Constraints: Insufficient funding or high costs can obstruct the adoption of new technologies or sustainable practices.

Infrastructure Limitations: Inadequate facilities or systems to support new projects or ideas.